Matched-Filter Noise Recovery and a Twelve-Member Sound-Event Ensemble for Indic Noise Event Detection

Saifuddin
Team syehsyoh; Senior Digital Transformation (AI & Automation) Expert; Datathon@IndoML 2026, Track 1
Code, heads and submissions: https://github.com/syehsyoh-dokling/NoiseDetection

***Abstract*—Track 1 asks for the onset and offset of every noise event in Vaani field recordings, scored by an event-based F1 (both boundaries within 20 % of the reference duration, at least 50 ms) plus a per-clip segment Dice. Our final submission (Codabench submission 994586) placed 5th of 82 teams with a combined score of 1.600895 (F1 0.780264, Dice 0.820631; maximum 2.0). The synthetic clips appear to be clean speech with noise pasted from a finite library; mixture minus the released clean reference recovers the pasted waveforms, and a normalised cross-correlation (NCC) matched filter over 1,198 such templates locates them in the test clips. Natural clips go to a frame-level sound-event detector (SED): a weighted mean of twelve bidirectional-GRU (BiGRU) heads over frozen or fine-tuned Whisper, AST, BEATs and Dasheng encoders, decoded with per-event peak-relative boundaries. Training uses the public Vaani Gold/Silver tiers and, for one member, pseudo-labels on unlabelled test audio. We report the lineage, out-of-fold component scores, compute cost, and the reproduction after the fine-tuned checkpoints were lost.**

I. Training Data and Preprocessing

Vaani data. The released validation set holds 2,501 clips (3.51 h, 16 kHz mono): 1,806 natural field recordings and 695 synthetic clips, each synthetic clip with its clean reference, from 28 states, 141 state/district pairs and 27 languages, with 5,157 labelled events in seven top-level categories (human non-speech 1,924; vehicle/traffic 1,281; animal 1,142; baby/child 496; phone/signal/alarm 188; singing/music 68; appliance/machine 58). The test set holds 5,517 unlabelled clips; no test labels were released or accessed.

External data (permitted by the rules [1], which allow additional public or proprietary data). We used the Gold tier of the Vaani noise-event dataset [2], [3] (10,772 clips, 19,414 verified-timestamp events; CC BY 4.0, gated access which we hold) for every member except wsA and wlP, and its Silver tier (unverified timestamps) for wlHS (14,529 clips from the Hugging Face release) and bFS (the first 12,000 clips of the Kaggle copy ayanokoji2028/vaani-noise-hf of [2], HF file names), at loss weight 0.4 (member tags: Table I). Overlap was checked by Vaani file stem, a unique clip id: Gold shares no clip with the test set (0 of 5,517) or the validation set (0 of 2,501) and no recording session (file-name prefix) with any test clip; 13 of 10,772 Gold clips share a session with a validation clip (negligible for OOF). Both Silver extractions exclude every validation and test file name (checked by name only, not by session). Speaker/district overlap with the test set was not checked (domain, not label, leakage).

Test-side information. (a) The leaderboard score of each upload was the final selection criterion; Codabench [1] lists 47 of our uploads (46 scored; 995069, 17 Oct, after the final, failed) and the first (921278, 10 Sep) is no longer listed, so at least 48 were made (quota 5/day, 100 total). (b) Member wlP was trained on pseudo-labels [4] that three validation-only Whisper heads (small, medium, large, weighted 1/1/2) produced on the 3,353 test clips without a library hit, a transductive use of unlabelled test audio (the organisers list semi-supervised use of unlabelled Vaani audio as a novelty direction). The fold-f student only sees labels from teachers that never saw fold f (leak-free OOF); Table III gives the ensemble without it.

Template library. Outside the pasted spans ($\pm$150 ms) the residual (mixture minus clean reference) of a synthetic validation clip is zero at most samples (median of per-clip medians 0, maximum 4.4×10^{-3} at full scale 1), so every labelled synthetic event cut from the residual is the pasted noise sample. Near-duplicates (NCC > 0.995) were merged, leaving 1,198 templates, each with its source clips for leave-one-clip-out (LOO) evaluation.

Features. On a 20 ms grid, the tokens of three or four layers per encoder are averaged over frequency (AST, BEATs, Dasheng), layer-normalised per frame, averaged over layers and interpolated onto the grid (Whisper frames are used directly). Targets are eight binary channels (any noise + seven categories): 24 % of natural reference events overlap in time with an event of another category, which the event-F1 term scores separately.

Validation protocol. Model selection used 5-fold cross-validation grouped by state/district (GroupKFold; 500–501 clips, 27–29 districts per fold) to mimic unseen test districts; every validation number is out-of-fold (OOF) with the test decoding, the library in LOO mode and cross-fitted null levels (two halves of the natural clips calibrate each other). Fusion weights, decode and matcher parameters were fitted on these OOF predictions (no nested CV), so OOF scores are optimistic for them.

TABLE I

THE TWELVE MEMBERS (TRAINING RECIPE IN SEC. II). V VALIDATION FOLDS, G GOLD, S SILVER (W 0.4), P PSEUDO-LABELS (W 0.5); W-S / W-L WHISPER-SMALL / LARGE-V3; FT = ENCODER FINE-TUNED (*PATCH EMBEDDING FROZEN); CROP IN FRAMES (FR) OR S. SEED 0, EXCEPT WLH2/WLH3 (1/2) AND BFS (7).

Tag	Encoder	Layers	Data	Ep/bs/crop	w
wsA	W-S	3,6,9,12	V	25/16/600 fr	0.5
wlP	W-L	8,16,24,32	V+P	25/16/600 fr	3
wlH/wlH2/wlH3	W-L	8,16,24,32	V+G	8/8/400 fr	2/6/1
wlHS	W-L	8,16,24,32	V+G+S	6/8/400 fr	6
aH/bH/dH	AST/BEATs/Dasheng	4,8,12	V+G	8/8/400 fr	8/8/4
aFG	AST ft (12 blocks)	4,8,12	V+G	3/8/10.24 s	8
bFG	BEATs ft*	4,8,12	V+G	3/8/10.24 s	8
bFS	BEATs ft*, no hold-out	4,8,12	all V+G+S	2/8/10.24 s	4

TABLE II

LEADERBOARD (LB) LINEAGE: 10 OF OUR ≥48 UPLOADS. OOF / ΔOOF: OUT-OF-FOLD SCORE / GAIN OVER THE PREVIOUS ROW (FOLDS UP OF 5), WHERE RECORDED. MED / THR: MEDIAN WINDOW (FRAMES) / THRESHOLD OF THE HUMAN-NON-SPEECH BRANCH. Q1 = FINAL CONFIGURATION WITHOUT BFS.

Upload	Change	OOF	ΔOOF	LB F1	LB Dice	LB comb.
921278[c]	first upload, 10 Sep	–	–	0.7113	0.7478	1.4591
924220	v14: 6 Whisper + 2 AST frozen members	1.4466	–	0.7752	0.8055	1.5808
929944	+ aFG:8, aA dropped[b]	–	–	0.7739	0.8130	1.5868
932046	+ bH:8	–	–	0.7741	0.8148	1.5890
932698	+ bFG:8, dH:4	–	–	0.7752	0.8187	1.5938
933670	boundary calibr. (med 11, thr 0.32)	–	+0.0024 (5)	0.7778	0.8194	1.5972
936587	med 13, thr 0.34	–	+0.0017 (4)	0.7799	0.8201	1.5999
991824	med 15	–	–	0.7802	0.8198	1.6000
991838	other-branch gap 2.0 (Q1)	1.4787	+0.0007 (4)	0.7806	0.8195	1.6002
994586	**+ bFS:4 (final)**	n/a[a]	–	**0.7803**	**0.8206**	**1.6009**

[a] bFS trains on all validation clips: no OOF estimate; weight set by a one-slot leaderboard probe. [b] aA, the second AST member of v14, dropped after its artifacts were lost (v14 without it: 1.4474 OOF). [c] No longer listed on Codabench; from our records. Differences below 0.001 between the last rows cannot be distinguished from leaderboard noise, which we did not quantify.

II. PROPOSED METHOD

Background. Frame-level CRNN classifiers [5], [6] are a standard approach to SED, evaluated with event- and segment-based metrics [7], and semi-supervised SED with unlabelled audio is a DCASE task [8]; Whisper's encoder keeps background-sound identity [9], and AudioSet [10] transformers (AST [11], BEATs [12], Dasheng [13]) are strong audio encoders. The synthetic half is detection of a known signal, as in audio fingerprinting [15], for which the matched filter is SNR-optimal in white Gaussian noise [16]; we use its normalised form, fast NCC [17].

Matched filter. For a test clip x and template T_i, the matcher computes the NCC $c_i(s)$ at every sample lag s with FFTs on the GPU and keeps the top 2 peaks per template and clip (0.3 s apart, NCC > 0.2). A per-template null level ν_i is the 0.999 quantile of $\max_s c_i(s)$ over the natural validation clips (unstored values count as 0.2, so the effective NCC threshold is ≥ 0.3). Candidates are kept when the margin $c_i(s) - \nu_i \geq 0.1$ and selected greedily by margin with at most 30 % mutual overlap. In LOO mode the matcher alone scores F1 0.9819 / Dice 0.9723 on the 695 synthetic clips (Table III); it fires on 687 of them and on 73 of the 1,806 natural clips.

SED ensemble. Each head is a BiGRU (hidden 192, dropout 0.3) with a linear layer to the 8 channels, trained with binary cross-entropy (masked to the clip's frames, per-channel weights), SpecAugment-style masking [18] and AdamW [19] (wd 10^{-2}) under a one-cycle schedule (peak 10^{-3}; ft encoders 2×10^{-5} with fp16 autocast and gradient checkpointing), last epoch kept; a member's test probability is the mean of its 5 fold heads (bFS: one model). Table I lists the members; the ensemble is $p = \sum_m w_m p_m / \sum_m w_m$. The weights of the eleven members with OOF predictions were set by coordinate descent over a coarse grid on the natural clips (`tune_weights.py`; the matcher dominates the synthetic ones); bFS's weight was a leaderboard probe. Members were added one at a time, each given one leaderboard upload ("one-slot probe") whose score then decided; per-member OOF records were not retained.

Decoding and fusion. Natural clips use two branches: $p_h = \min(p_{\text{hns}}, p_0)$ for human non-speech (hns) and $p_o = \min(\max_{c\neq\text{hns}} p_c, p_0)$ for the other categories, p_0 being the any-noise channel. Each branch is decoded peak-relatively: a region is detected where the median-filtered curve (window 15 / 7 frames) clears a floor of 0.2, is kept if its peak reaches 0.34 / 0.30, and both boundaries are placed where the curve crosses α times that event's own peak ($\alpha = 0.6$ / 0.5); introduced before upload 924220, this was worth +0.029 natural OOF event F1 over one global threshold on the same probabilities. Events shorter than 0.1 s are dropped, gaps below 0.1 s / 2.0 s are closed, and the branches are merged with duplicate removal at IoU 0.5. A clip with any library hit is treated as synthetic: its events are the template events plus any SED event (any-noise channel, threshold 0.7, median 3, minimum 0.1 s, gap 0.04 s) that does not overlap them (`models/fusion_Q1_bFSadd4.json`). The system is offline and non-causal: the heads are bidirectional, the decoder needs the whole event, the encoders see the whole clip.

Compute and software. Python 3.13.6, PyTorch 2.14.0 (CUDA 12.6). Frozen heads (on cached float16 features), all aFG folds and bFG folds 0–2 were trained on one RTX 4060 Laptop GPU (8 GB); bFG folds 3–4 and bFS on Kaggle T4 GPUs. On one T4 per job (retrain logs): one aFG fold 64–67 min, one bFG fold about 55 min, bFS 75 min; one fine-tuned encoder pass over the 5,517 test clips (10.81 h of audio) 4.5–5 min (an estimated 55 min for the eleven); the five frozen-backbone passes plus downloads under 1.4 h; the matched filter 84.6 min on the laptop GPU (FFT up to 2^{19}). Encoders: openai/whisper-small, whisper-large-v3 [20] (Apache 2.0), MIT/ast-finetuned-audioset-10-10-0.4593 [11] (BSD 3), BEATs iter3+ AS2M [12] (MIT), mispeech/dasheng-base [13] (Apache 2.0); not redistributed, fine-tuned derivatives keep the notices.

III. RESULTS AND CONCLUSION

Leaderboard. Table II traces the system from the first upload (1.4591) to the final 1.6009 (our highest listed upload). The changes up to v14 (more members, Gold data, the split

TABLE III
OOF SCORES OF Q1 ON VALIDATION: COMBINED (F1 / DICE). OVER THE 5 FOLDS (MEAN ± SD, RANGE): Q1 1.476±0.077 (1.379–1.590), NO WLP 1.474±0.075 (1.378–1.583), SED ONLY 1.201±0.031 (1.169–1.243). "RETRAINED FT": Q1 WITH THE RETRAINED AFG/BFG.

System	All	Natural	Synthetic
Q1 full	1.4787 (.689/.789)	1.1570 (.447/.711)	1.9868 (.993/.994)
no wlP	1.4759 (.687/.789)	1.1527 (.443/.710)	1.9866 (.992/.994)
retrained ft	1.4771 (.689/.789)	1.1542 (.445/.709)	1.9869 (.993/.994)
SED only	1.2005 (.450/.751)	1.1409 (.433/.708)	1.3378 (.476/.862)
Templates	0.8871 (.598/.290)	0.0691 (.042/.027)	1.9542 (.982/.972)

TABLE IV
NATURAL CLIPS, Q1 OOF: RECALL OF THE 2,944 REFERENCE EVENTS PER CATEGORY UNDER THE METRIC'S TOLERANCE (PREDICTIONS CARRY NO CATEGORY).

	human	vehicle	animal	baby	alarm	music	machine	all
n	1,254	655	594	265	110	40	26	2,944
recall	0.622	0.282	0.269	0.215	0.300	0.225	0.231	0.418
med. dur. (s)	0.32	0.70	0.79	0.70	0.60	2.78	6.23	–

and peak-relative decodes) are not separable in our records and give most of the gain (to 1.5808); fine-tuning AST, dropping aA and adding BEATs and Dasheng add +0.0131; OOF-tuned decoding changes +0.0063; bFS +0.0007, below what we can distinguish from noise. The gap to rank 1 (1.6173) is 0.0164: F1 0.0092, Dice 0.0072. Selection was leaderboard-guided, so the final score carries an unquantified optimistic bias. OOF and LB scores of one configuration differ by 0.12 (1.4787 vs 1.6002); we hypothesise (untested) that this comes from the test set being 39 % synthetic-like (2,164 of 5,517 clips with a library hit, against 28 % truly synthetic and 30 % with a LOO hit in validation) and from the test run using the full library.

Components and errors. Table III: on synthetic clips the matcher alone gives 1.9542, the SED alone 1.3378; on natural clips the SED alone 1.1409, the matcher alone 0.0691, the fusion +0.016 over SED. wlP is worth +0.0028 pooled; the paired fold difference is positive on 4 of 5 folds (mean +0.0026, $t \approx 1.7$, not significant); its leaderboard effect, where its pseudo-labels come from the scored clips, was not measured. Table IV (recall only): 62 % of human non-speech events (median 0.32 s) are recovered but only 21–30 % of the other categories; boundary ambiguity of long, stationary noises and their scarcity are plausible causes we did not separate (on a pre-v14 ensemble 12.9 % of natural events were truly missed, 52 % failed a boundary tolerance). Tried and rejected: a soft-Dice loss term (wlHSD, refuted out-of-fold), a stacker, CLAP and CED [14] members, a fine-tuned Whisper (wFG), a fine-tuned Dasheng (dFG) and a second AST seed (configurations in `models/fusion_*.json`; reasons not uniformly recorded).

Reproduction. After the final upload we lost the backbones, fine-tuned checkpoints, cached test probabilities and test template candidates; the 45 frozen fold heads, configurations, library and scored zip survived. We retrained aFG, bFG and bFS with the same scripts and seeds on Kaggle T4s (the originals of aFG and bFG folds 0–2 ran on the laptop), reran the matcher and recomputed the frozen members from the released heads. The retrained aFG/bFG give 1.4771 OOF (vs 1.4787); the rebuilt submission flags the identical set of 2,164 synthetic-like clips, is identical on 41.8 % of clips (2,008 of 2,164 synthetic-like, 298 of 3,353 natural-like; 13,442 vs 13,451 events) and scores 1.9858 (F1 0.9943 / Dice 0.9915) with the scored file as reference. The rebuilt SED probabilities (different GPU) are thus not bit-identical; the released fine-tuned checkpoints (eleven, 3.96 GB, release assets) are the retrained ones, and bFS (no OOF) rests on this agreement. wlP's pseudo-labels and teacher heads are not released: its five heads make inference, not retraining, reproducible.

Limitations and conclusion. The matcher needs the test set to reuse the validation noise library (else the SED branch, 1.2005 OOF, carries the score); the ensemble is large, offline and non-causal; wlP is transductive; natural-clip F1 (0.45) is the weakest part. Known-signal detection plus per-event SED boundary placement gave a 5th-place system; code, configurations, all heads and the retrained checkpoints are released (wlP's pseudo-labels excepted).

REFERENCES

[1] Datathon@IndoML 2026, "Track 1: Noise event detection," https://www.codabench.org/competitions/17825, accessed Oct. 2026.

[2] ARTPARK-IISc, "Vaani noise event dataset," 2026, https://huggingface.co/datasets/ARTPARK-IISc/Vaani-Noise-Event-Dataset.

[3] S. Pulikodan *et al.*, "VAANI: Capturing the language landscape for an inclusive digital India," arXiv:2603.28714, 2026.

[4] D.-H. Lee, "Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks," in *ICML Workshop on Challenges in Representation Learning*, 2013.

[5] E. Çakır *et al.*, "Convolutional recurrent neural networks for polyphonic sound event detection," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 25, no. 6, pp. 1291–1303, 2017.

[6] A. Mesaros *et al.*, "Sound event detection: A tutorial," *IEEE Signal Process. Mag.*, vol. 38, no. 5, pp. 67–83, 2021.

[7] A. Mesaros, T. Heittola and T. Virtanen, "Metrics for polyphonic sound event detection," *Applied Sciences*, vol. 6, no. 6, 162, 2016.

[8] N. Turpault *et al.*, "Sound event detection in domestic environments with weakly labeled data and soundscape synthesis," in *Proc. DCASE Workshop*, 2019.

[9] Y. Gong, S. Khurana, L. Karlinsky and J. Glass, "Whisper-AT: Noise-robust automatic speech recognizers are also strong general audio event taggers," in *Proc. Interspeech*, 2023.

[10] J. F. Gemmeke *et al.*, "Audio Set: An ontology and human-labeled dataset for audio events," in *Proc. ICASSP*, 2017.

[11] Y. Gong, Y.-A. Chung and J. Glass, "AST: Audio spectrogram transformer," in *Proc. Interspeech*, 2021.

[12] S. Chen *et al.*, "BEATs: Audio pre-training with acoustic tokenizers," in *Proc. ICML*, 2023.

[13] H. Dinkel *et al.*, "Scaling up masked audio encoder learning for general audio classification," in *Proc. Interspeech*, 2024.

[14] H. Dinkel, Y. Wang, Z. Yan, J. Zhang and Y. Wang, "CED: Consistent ensemble distillation for audio tagging," in *Proc. ICASSP*, 2024.

[15] A. Wang, "An industrial-strength audio search algorithm," in *Proc. ISMIR*, 2003.

[16] G. L. Turin, "An introduction to matched filters," *IRE Trans. Inf. Theory*, vol. 6, no. 3, pp. 311–329, 1960.

[17] J. P. Lewis, "Fast normalized cross-correlation," in *Vision Interface*, 1995.

[18] D. S. Park *et al.*, "SpecAugment: A simple data augmentation method for automatic speech recognition," in *Proc. Interspeech*, 2019.

[19] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019.

[20] A. Radford *et al.*, "Robust speech recognition via large-scale weak supervision," in *Proc. ICML*, 2023.